NOTE The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed but will be changed the next time this document is revised.

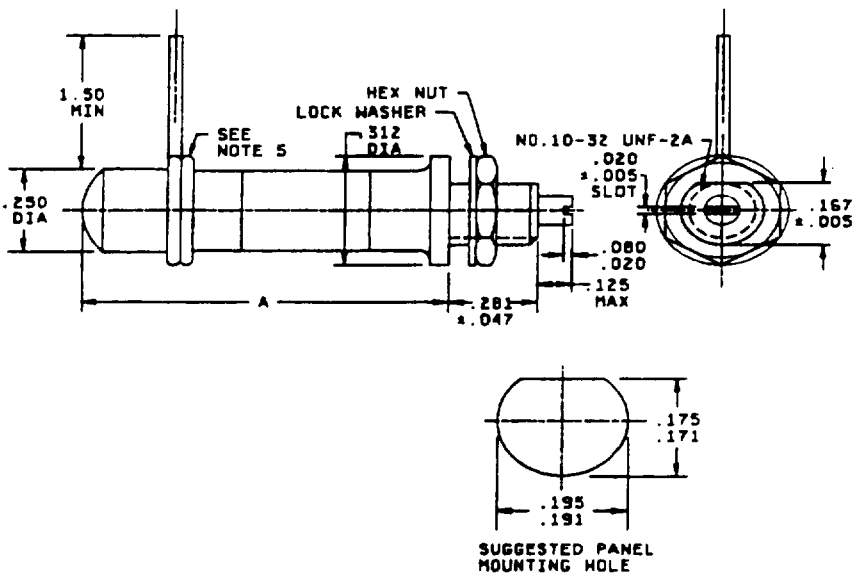
MIL-PRF-14409/4B
5 October 1970
SUPERSEDING
MIL-C-14409/4A
9 June 1969

PERFORMANCE SPECIFICATION SHEET

# CAPACITORS, VARIABLE (PISTON TYPE, TUBULAR TRIMMER), STYLES PC38 AND PC48

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-PRF-14409.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.




| INCHES | MM |
|---|---|
| .005 | .13 |
| .016 | .41 |
| .020 | .51 |
| .047 | 1.19 |
| .080 | 2.03 |
| .125 | 3.18 |
| .167 | 4.24 |
| .171 | 4.34 |
| .175 | 4.45 |
| .191 | 4.85 |
| .195 | 4.95 |
| .250 | 6.35 |
| .281 | 7.14 |
| .312 | 7.92 |
| 1.500 | 38.10 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.016.
4. The flat on the mounting bushing shall extend to the mounting surface of the shoulder.
5. There shall be a minimum of one and one-half turns of wire around the body. These turns of wire shall be completely soldered to the body. For wire sizes see table I.
6. For dimension A see table I.

(B) denotes changes

FSC 5910



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REQUIREMENTS

Dimensions: See figure and table I.

DC voltage rating: See table I.

Construction: Sealed glass or quartz, see figure and table I.

Capacitance range: See table I.

Insulation resistance (IR): Method 302 of MIL-STD-202, test condition B (500 volts ± 10 percent), not less than 1,000,000 megohms, except characteristics J and H shall be not less than 100,000 megohms at 125°C.

Quality factor (Q): See table I.

Temperature cycling: Method 102 of MIL-STD-202, test condition D, except the step 3 temperature shall be 125°C ± 5°C for characteristics J and H, and 150°C ± 5°C for characteristic Q.

Immersion: Method 104 of MIL-STD-202, test condition A.

Seal test: Applicable.

TABLE I. Type designation and characteristics.

| Type designation | DC rated voltage | Capacitance range | | Dimension 1/ | Q | Wire size AWG | Dielectric material |
|---|---|---|---|---|---|---|---|
| | | Min | Max | A ± .031 (.79) | | | |
| | volts | pf | pf | | | | |
| PC38Q1R5 | 750 | 0.6 | 1.8 | .359( 9.12) | 1,500 | 24 | Quartz |
| (B) PC38Q5R5 | 1,250 | 0.6 | 5.5 | .625(15.88) | 1,500 | 22 | Quartz |
| PC38Q9R5 | 1,250 | 0.6 | 9.5 | 1.062(26.97) | 1,500 | 22 | Quartz |
| PC38Q160 | 1,250 | 0.8 | 16.0 | 1.656(42.06) | 1,500 | 22 | Quartz |
| PC48J5R5 | 750 | 0.6 | 5.5 | .359( 9.12) | 1,000 | 24 | Glass |
| PC48J110 | 1,250 | 0.6 | 11.0 | .594(15.09) | 900 | 22 | Glass |
| PC48H160 | 1,250 | 0.6 | 16.0 | .812(20.62) | 800 | 22 | Glass |
| PC48H230 | 1,250 | 0.8 | 23.0 | 1.062(26.97) | 700 | 22 | Glass |
| PC48H380 | 1,250 | 1.0 | 38.0 | 1.656(42.06) | 500 | 22 | Glass |

1/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm. Millimeters are in parentheses.

TABLE II. Substitutability data.

| Superseded type designation in MIL-C-14409/4 | Substitutable type designation in MIL-C-14409/4A |
|---|---|
| PC38J4R5 | PC48J5R5 |
| PC38J8R5 | PC48J110 |
| PC38H120 | PC48H160 |
| PC38H180 | PC48H230 |
| PC38H300 | PC48H380 |

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Custodians
Army - EL
Navy - EC
Air Force - 11

Review activities
Army -
Navy - AS, OS
Air Force -
DSA - ES

User activities
Army -
Navy - CG, MC
Air Force -

Preparing activity
Army - EL

(Project 5910-1010)



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